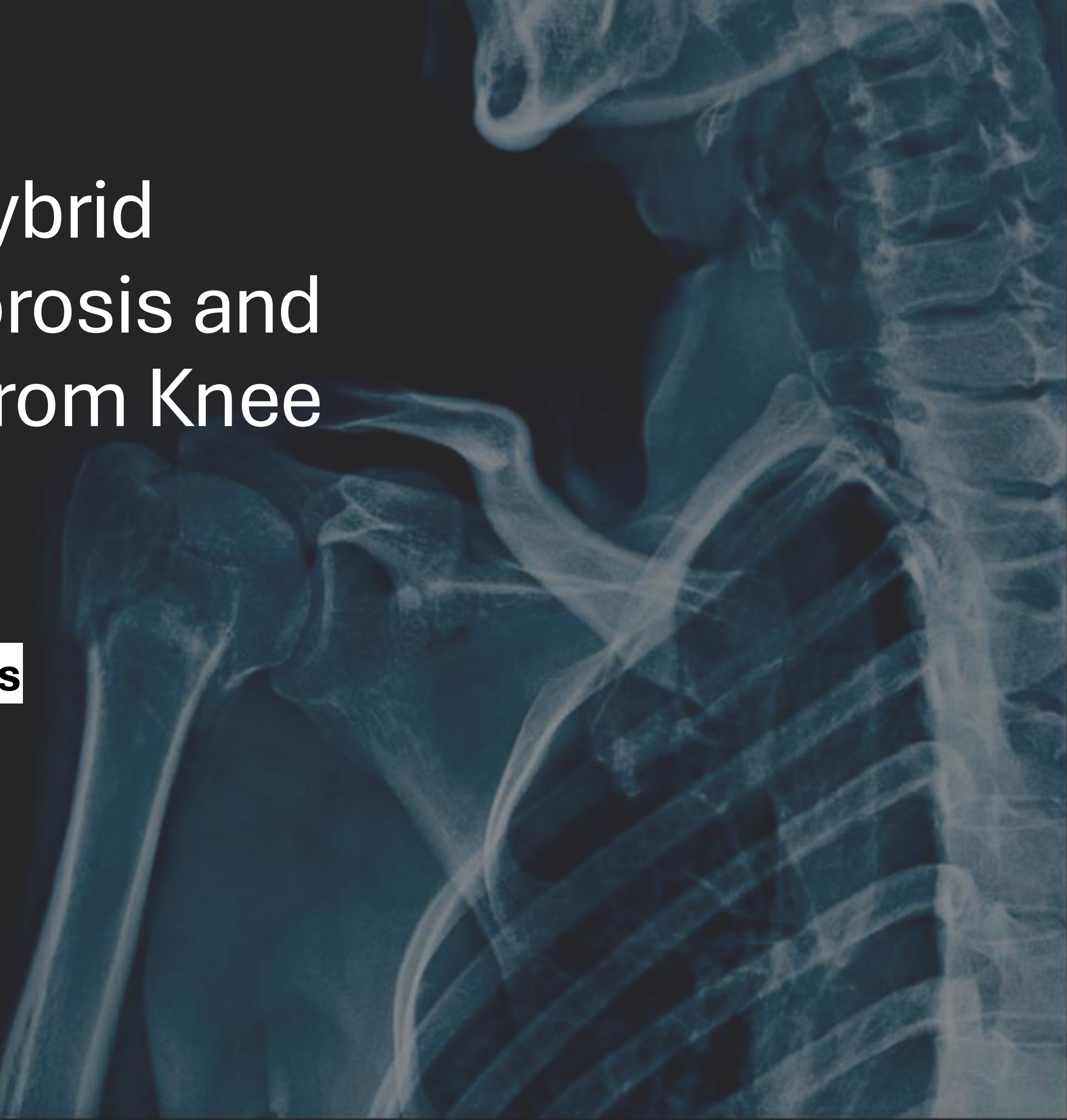

A Region-Aware Hybrid Framework for Osteoporosis and Osteopenia Detection from Knee X-ray Images

Smart Diagnostics for Stronger Bones

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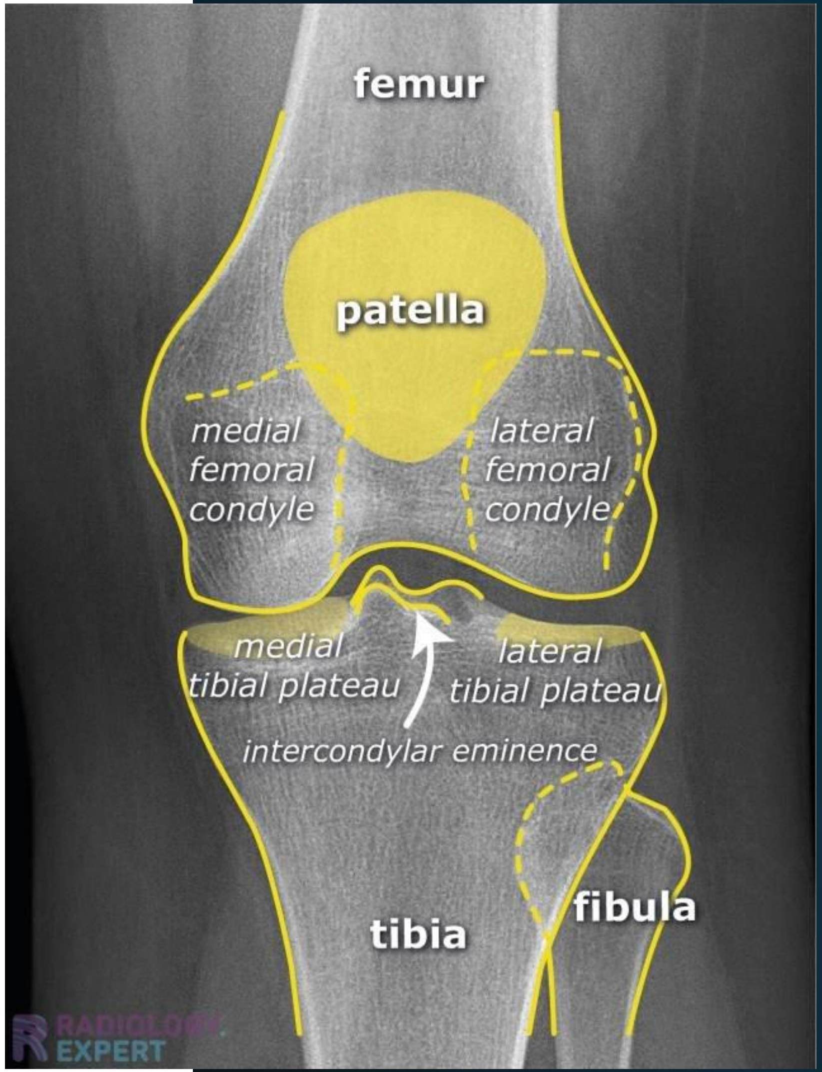
Problem Statement

Osteoporosis detection is currently a manual and time-consuming process that requires expert radiologists for accurate diagnosis. Most existing systems analyze the entire knee X-ray as a single unit, which reduces precision because different bones are affected differently. Therefore, there is a need for an automated and accurate system that can perform bone-specific analysis, enabling separate and more reliable diagnosis for individual regions like the femur and tibia.

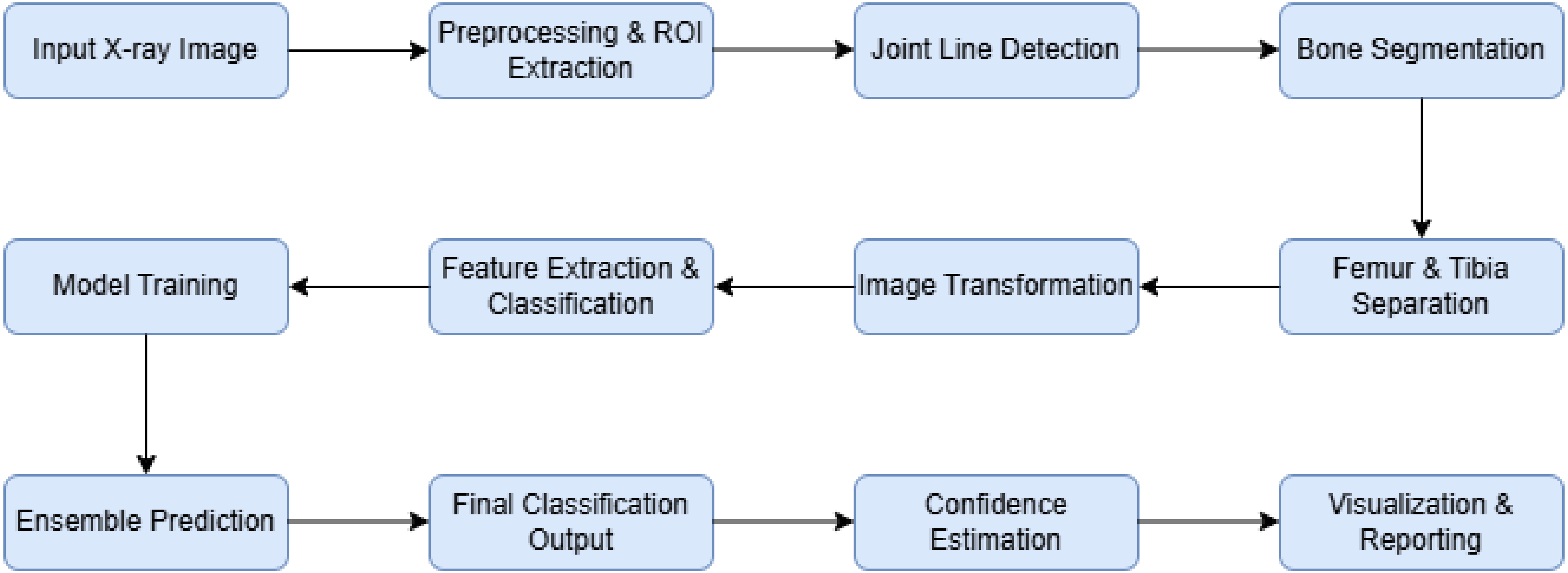


Proposed Solution

We developed a Region-Aware Hybrid Framework using knee X-ray images by combining image processing techniques and the EfficientNetV2-S deep learning model.

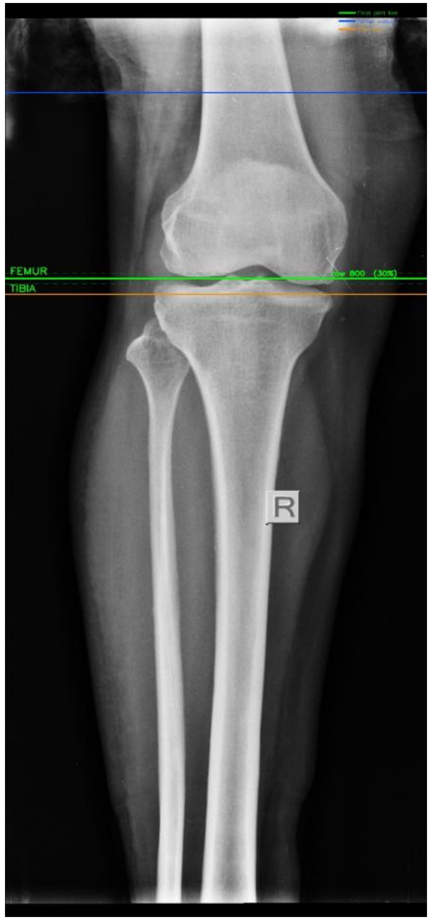
- First, image processing is used to segment the knee X-ray into femur and tibia regions
- Then, each region is analyzed separately to capture localized bone changes
- EfficientNetV2-S is used for feature extraction and classification
- This hybrid approach improves detection accuracy
- Uses low-cost X-ray images instead of DEXA scans, making it more accessible

Pipeline


Input X-ray Image
Preprocessing & ROI Extraction
Joint Line Detection
Bone Segmentation
Femur & Tibia Separation
Image Transformation
Feature Extraction & Classification
Model Training
Ensemble Prediction
Final Classification Output
Confidence Estimation
Visualization & Reporting

Demo Slide : Splitting an input image into femur and tibia

Whole bone image



Annotated image

Femur

Tibia

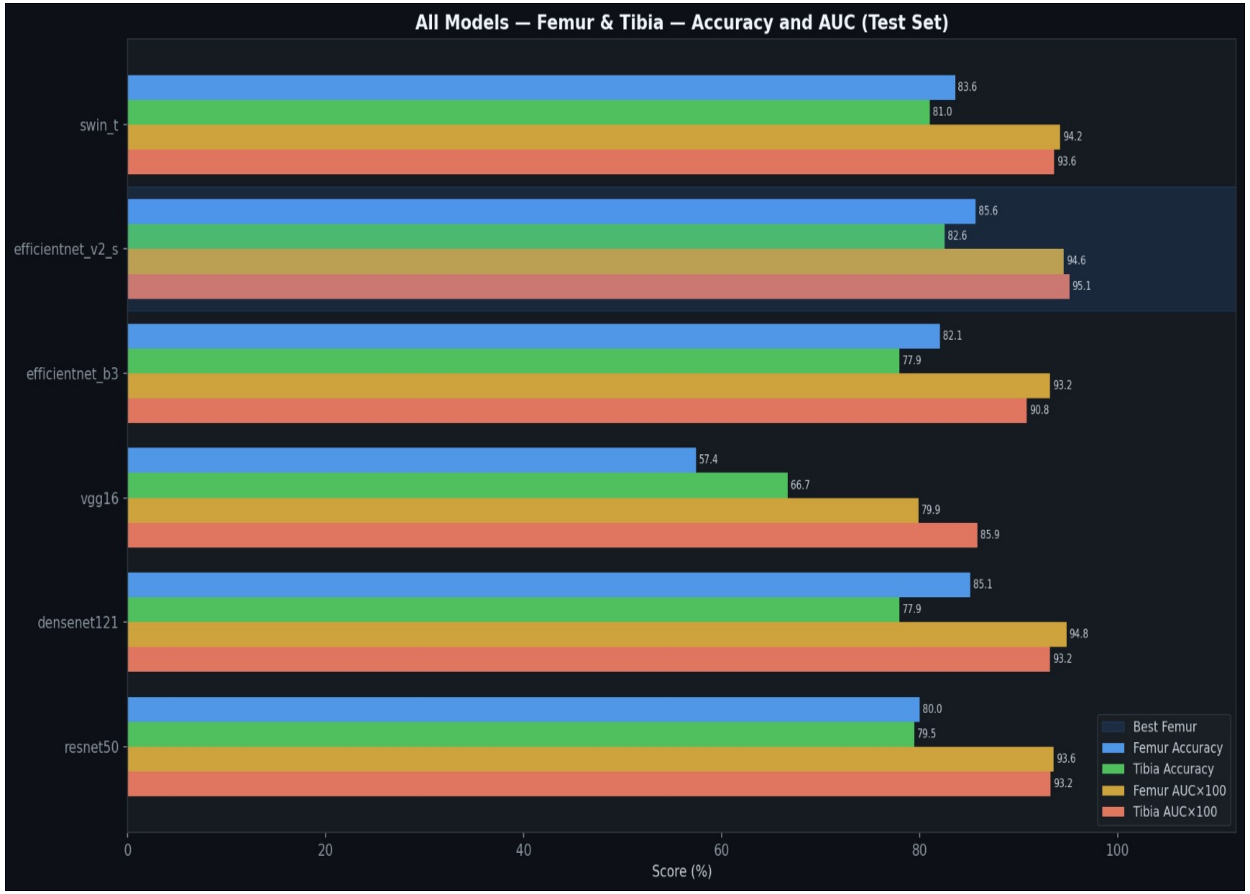
Results:

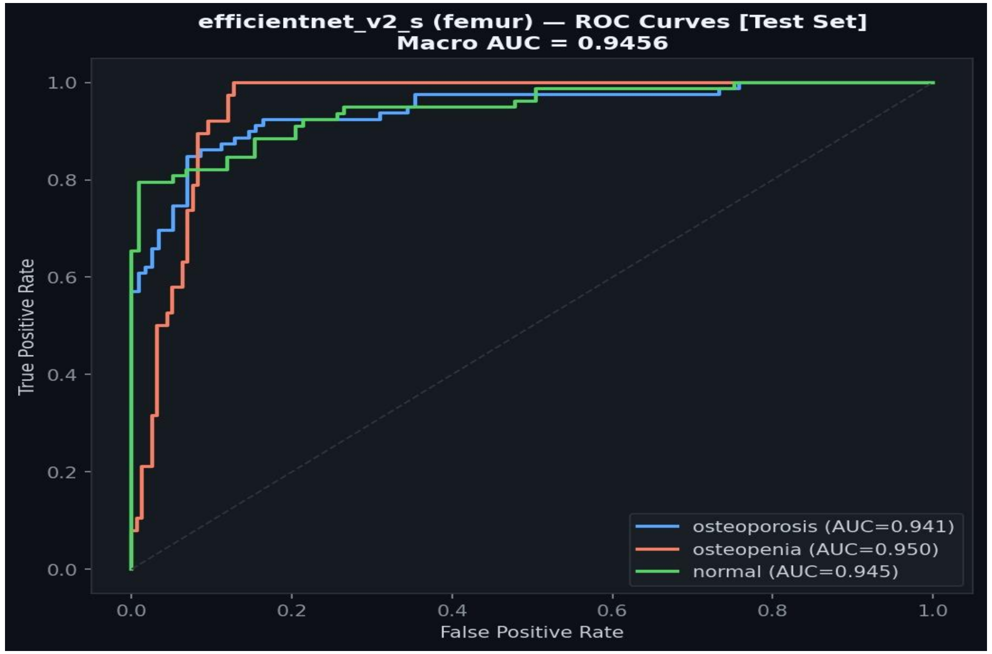
FEMUR RESULTS (ranked by accuracy)

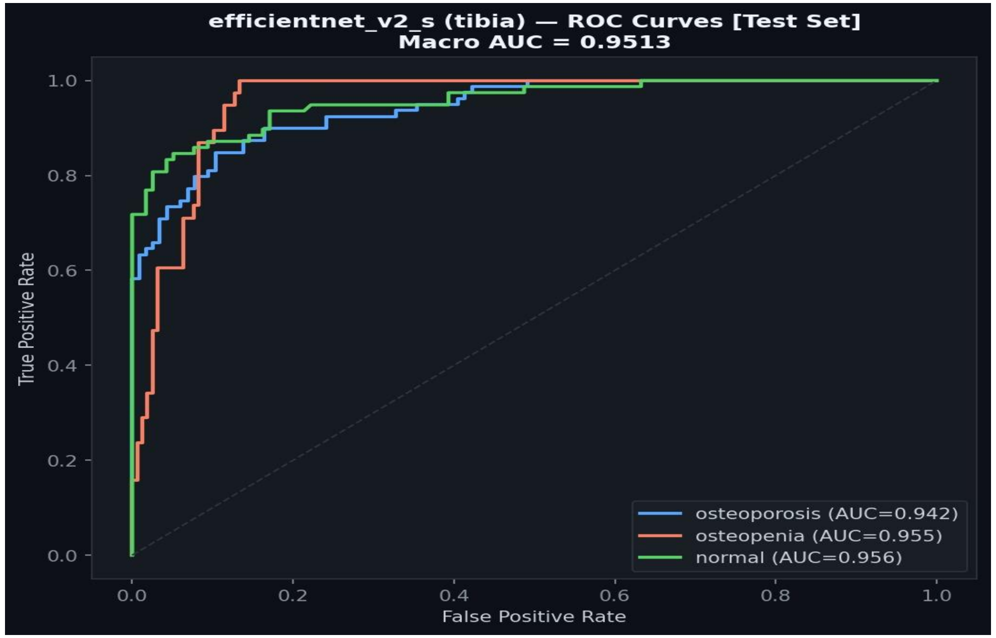
Rank	Model	Accuracy	Bal Acc	AUC	Time	
1	efficientnet_v2_s	85.64%	88.10%	0.9456	0.0m	← BEST
2	densenet121	85.13%	85.87%	0.9483	0.0m	
3	swin_t	83.59%	85.05%	0.9418	0.0m	
4	efficientnet_b3	82.05%	82.87%	0.9319	0.0m	
5	resnet50	80.00%	78.45%	0.9356	0.0m	
6	vgg16	57.44%	55.32%	0.7987	0.0m	

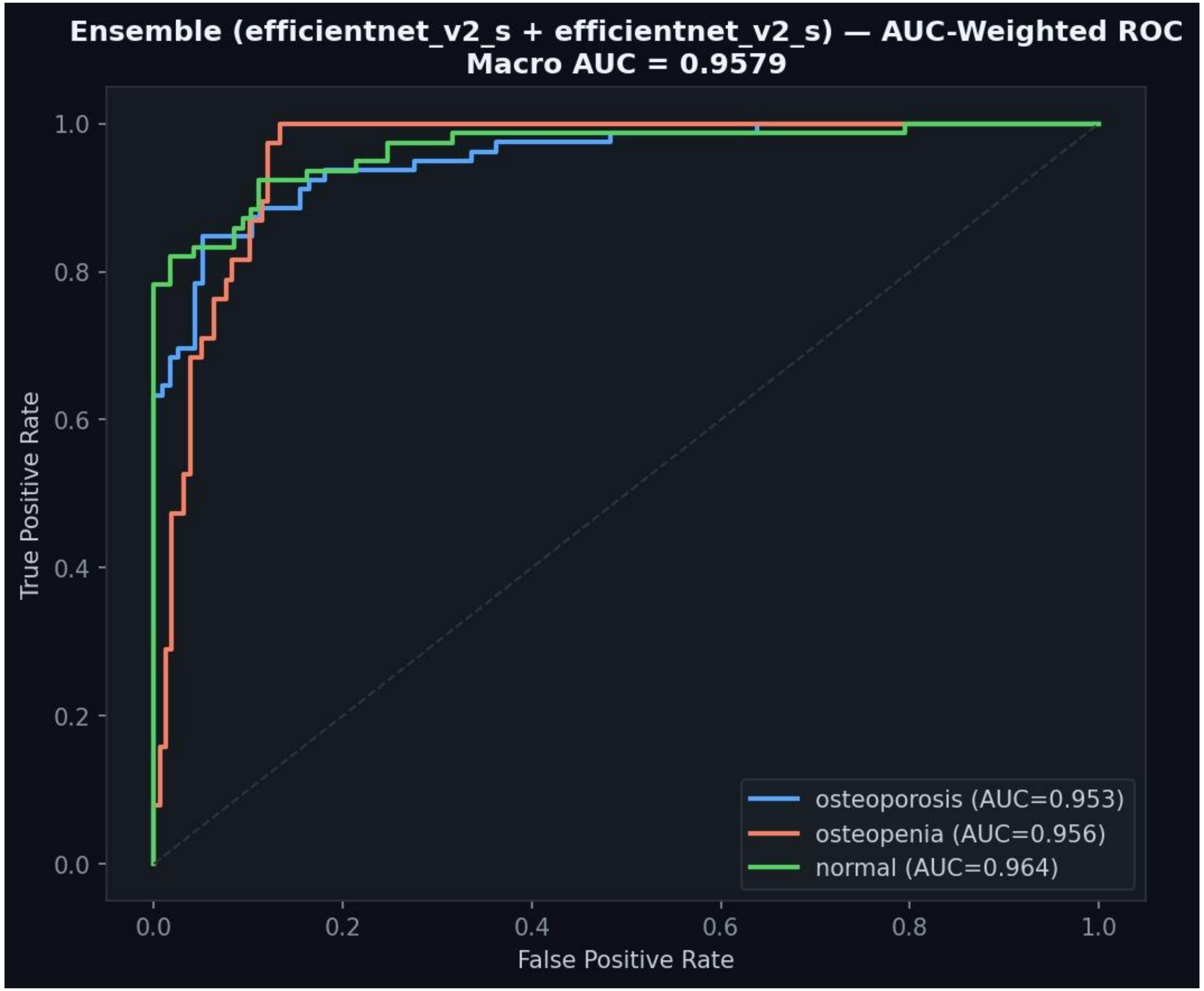
TIBIA RESULTS (ranked by accuracy)

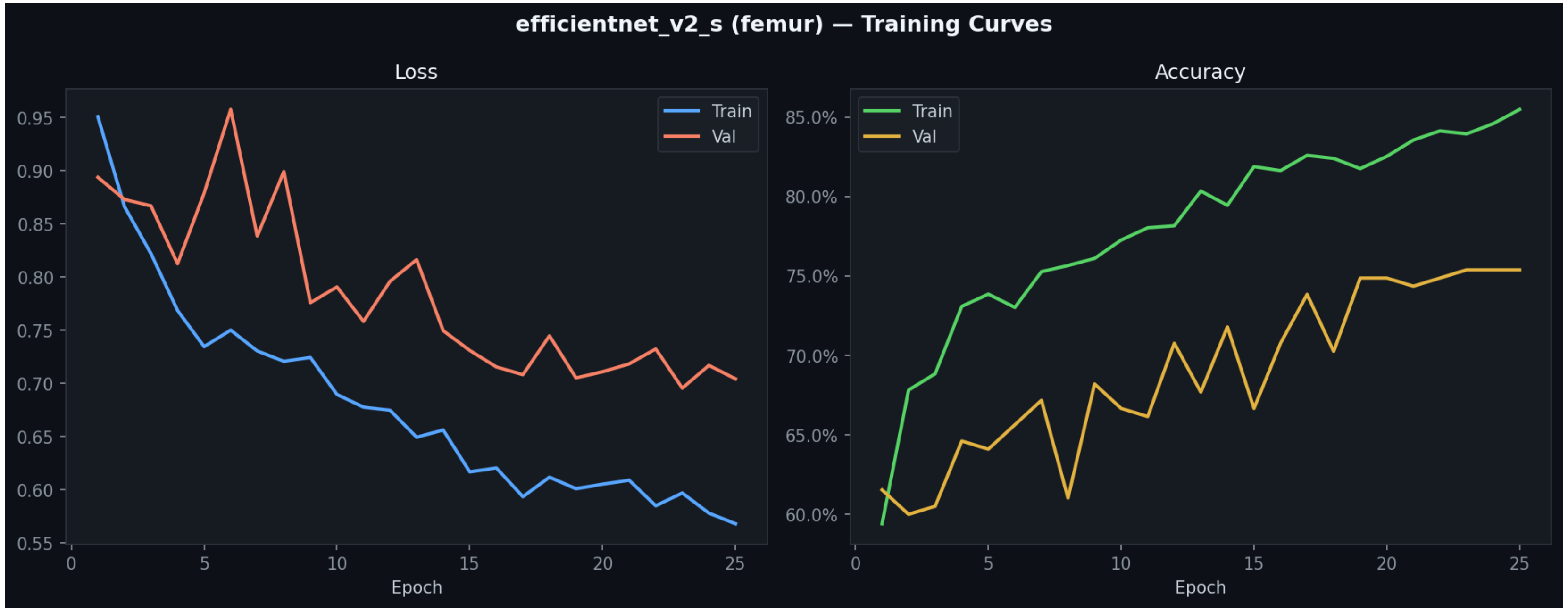
Rank	Model	Accuracy	Bal Acc	AUC	Time	
1	efficientnet_v2_s	82.56%	84.24%	0.9513	346.8m	← BEST
2	swin_t	81.03%	84.30%	0.9358	415.9m	
3	resnet50	79.49%	81.21%	0.9323	502.1m	
4	densenet121	77.95%	76.31%	0.9319	317.6m	
5	efficientnet_b3	77.95%	81.29%	0.9085	261.3m	
6	vgg16	66.67%	66.09%	0.8587	1197.4m	

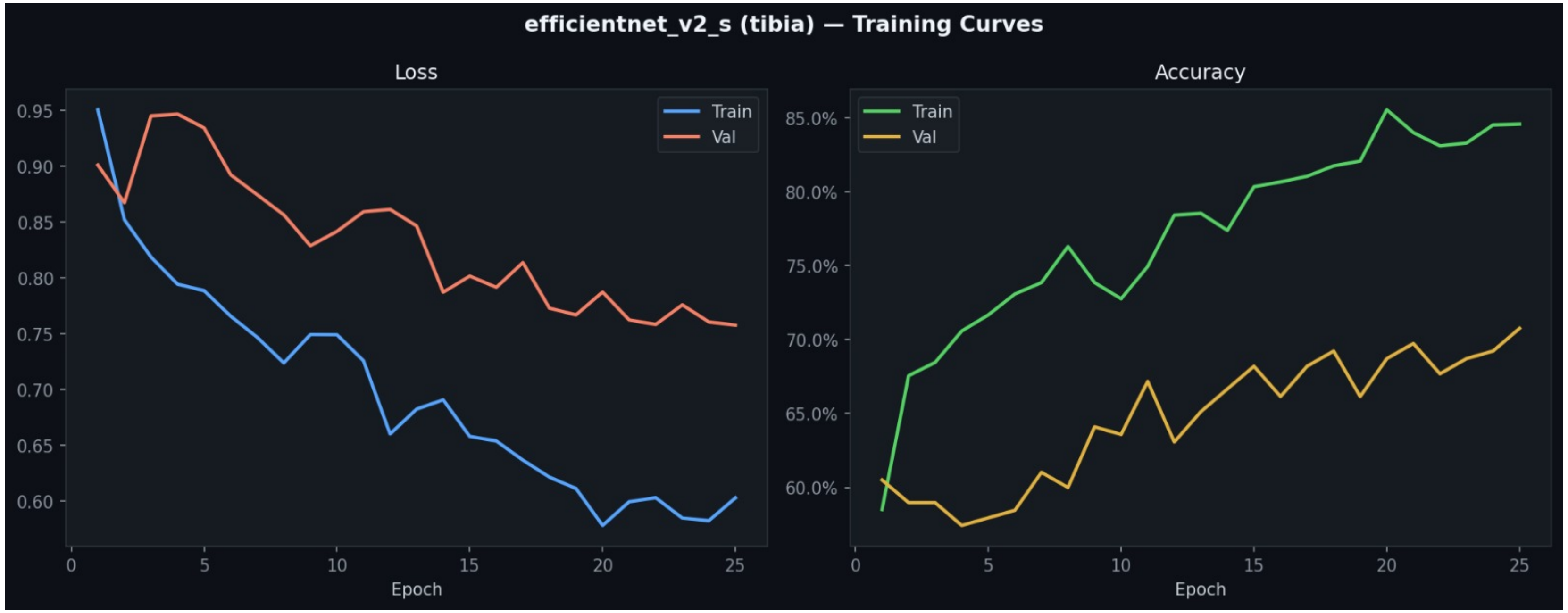



efficientnet_v2_s (femur) — ROC Curves [Test Set]
Macro AUC = 0.9456
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
osteoporosis (AUC=0.941)
osteopenia (AUC=0.950)
normal (AUC=0.945)


efficientnet_v2_s (tibia) — ROC Curves [Test Set]
Macro AUC = 0.9513
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
osteoporosis (AUC=0.942)
osteopenia (AUC=0.955)
normal (AUC=0.956)


Ensemble (efficientnet_v2_s + efficientnet_v2_s) — AUC-Weighted ROC
Macro AUC = 0.9579
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
osteoporosis (AUC=0.953)
osteopenia (AUC=0.956)
normal (AUC=0.964)

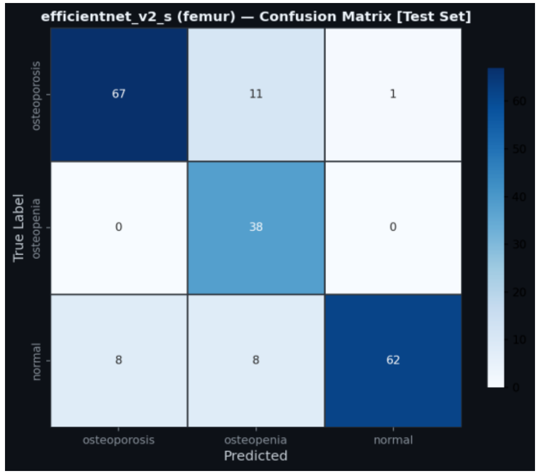

efficientnet_v2_s (femur) — Training Curves
Loss
Accuracy
Train
Val
Epoch
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
85.0%
80.0%
75.0%
70.0%
65.0%
60.0%
0
5
10
15
20
25

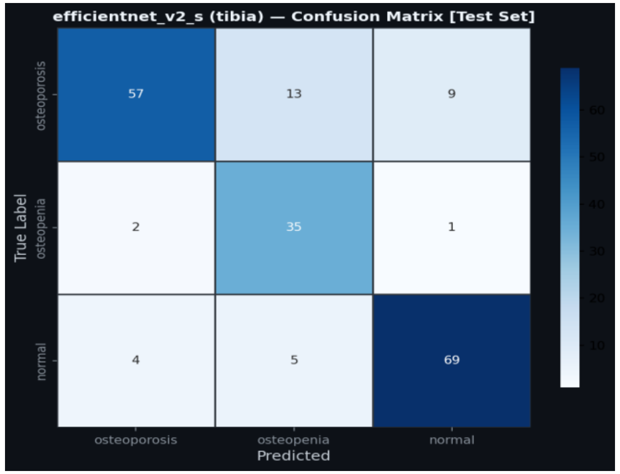

efficientnet_v2_s (tibia) — Training Curves
Loss
Accuracy
Train
Val
Epoch
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
85.0%
80.0%
75.0%
70.0%
65.0%
60.0%
0
5
10
15
20
25

```
BEST MODEL RECOMMENDATION FOR ENSEMBLE

Best Femur Model : EFFICIENTNET_V2_S
  Accuracy       : 85.64%
  Balanced Acc   : 88.10%
  ROC-AUC        : 0.9456
  Saved at       : models\efficientnet_v2_s_femur.pth

Best Tibia Model : EFFICIENTNET_V2_S
  Accuracy       : 82.56%
  Balanced Acc   : 84.24%
  ROC-AUC        : 0.9513
  Saved at       : models\efficientnet_v2_s_tibia.pth
```






```
Femur model   : EFFICIENTNET_V2_S  (AUC weight = 0.9456  [4
Tibia model   : EFFICIENTNET_V2_S  (AUC weight = 0.9513  [5
Formula       : prob = (50% × femur_prob) + (50% × tibia_pr

ENSEMBLE PERFORMANCE

Overall Accuracy     : 84.62%
Balanced Accuracy    : 86.82%
Macro ROC-AUC        : 0.9579

Per-Class Accuracy:
  osteoporosis       : 79.75%
  osteopenia         : 97.37%
  normal             : 83.33%

Classification Report:
              precision    recall  f1-score   support

osteoporosis       0.91      0.80      0.85        79
  osteopenia       0.66      0.97      0.79        38
      normal       0.93      0.83      0.88        78

    accuracy                           0.85       195
   macro avg       0.83      0.87      0.84       195
weighted avg       0.87      0.85      0.85       195
```

CONFIDENCE PREDICTION

Osteoporosis image

```
⚠ FINAL RESULT
Diagnosis : OSTEOPOROSIS
Risk      : HIGH RISK
Confidence: 96.1%

Per-bone confidence for 'osteoporosis':
Femur : 95.8%
Tibia : 96.4%

Per-bone breakdown:
Class                 Femur    Tibia    Ensemble
------------------------------------------------
osteoporosis          95.8%    96.4%       96.1%
osteopenia             1.8%     1.4%        1.6%
normal                 2.4%     2.2%        2.3%
```

Ablation Study

Model Variant	Accuracy (%)	Balanced Accuracy (%)	AUC
Femur only (EfficientNetV2-S)	85.64	88.10	0.9456
Tibia only (EfficientNetV2-S)	82.56	84.24	0.9513
Ensemble (50 / 50)	84.62	86.82	0.9578
Ensemble (AUC-weighted)	84.62	86.82	0.9579